Tanmay Bhuskute

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SUMMARY

Software Engineer and M.S. Computer Science candidate (Dec 2026) with two years at Siemens Industry Software automating, upgrading, and troubleshooting Teamcenter X PLM production environments across Linux, AWS, Kubernetes, and databases.

EDUCATION

Master of Science in Computer Science Aug 2024 - Dec 2026
University of Cincinnati
Relevant Coursework: Advanced Algorithms, Deep Learning, Intelligent Data Analysis, Database Theory.

Bachelor of Engineering in Computer Engineering Aug 2018 - Jul 2022
AISSMS College of Engineering
Relevant Coursework: Data Mining, Data Analytics, Machine Learning, Object-Oriented Programming.

TECHNICAL SKILLS

Programming: Java, Python, JavaScript, TypeScript, C/C++, SQL, Go, PowerShell.

Web & Backend: React, Node.js, Django, Flask, FastAPI, HTML/CSS, REST APIs.

Databases: PostgreSQL, MySQL, Oracle DB, DynamoDB.

Developer Tools: Teamcenter X, Linux, Docker, Kubernetes, AWS, Ansible, Terraform, ArgoCD, Helm, GitLab CI/CD, Jenkins, HashiCorp Vault, Git, Bash, SonarQube.

AI & Data: Gemini, Vertex AI, PyTorch, TensorFlow, scikit-learn, LangChain, pandas, NumPy.

WORK EXPERIENCE

Graduate Researcher | University of Cincinnati Feb 2026 – Present

- Designed memory-augmented LLM pipeline for Android GUI bug reproduction, evaluated against **ViBR** baseline across 53 runs on 20+ apps, achieving **25.7x latency reduction** (39.4s vs 1012.1s).

Software Development Engineer | Siemens Industry Software Jul 2022 – Jun 2024

- Rewrote legacy Ruby AWS environment creation in Python (**Boto3**) with error handling, automating **Teamcenter X** installation and linking across EC2, EFS, RDS, VPC, Route 53, and ALB, cutting deployment time by **40%**.
- Executed **44 production upgrades** in 4 months, moving Teamcenter X base platform, **Dispatcher**, and customized environments from 5.x, 6.x, and 7.x to version 8, including database schema upgrades, DSS Vault linking, and EC2 migration.
- Diagnosed and resolved live issues across Linux, EC2, databases, and Teamcenter X services using logs and metrics.
- Designed **Ansible-based Blue-Yellow-Green deployment** framework validating upgrades on a replica before cutover, cutting upgrade downtime by **50%** with backup-based rollback.
- Migrated **40+ production environments** from process-based Teamcenter X 8.x to containerized 9.x on Amazon EKS using ArgoCD, Helm, GitLab CI/CD, Amazon EFS, and HashiCorp Vault for secrets.
- Served as Quality Coach and SonarQube Specialist, increasing automated code coverage from **35% to 78%**; mentored **5 software engineers** through structured onboarding.

Software Developer Intern | Siemens Industry Software Mar 2022 – Jun 2022

- Automated provisioning of AWS resources with **Terraform**, reducing deployment time by **30%**.
- Developed **Go + Terratest** framework, catching **95%** of misconfigurations pre-production.

Software Developer Intern | Government of Maharashtra, Water Resource Department Sep 2021 – Dec 2021

- Built full-stack **React and MySQL** application digitizing departmental records and deployed on-premises MySQL with authentication and permissions, cutting document retrieval time by **90%** (5 min to 30 sec).

PROJECTS AND RESEARCH

NYC Mobility Forecasting [Docker, FastAPI, PostgreSQL, Redis, PySpark] Jul 2026

- Built ML system forecasting NYC taxi/rideshare demand across **263 zones** (**2.3M-row** hourly dataset); designed two-stage Poisson hurdle model improving MAPE **18.77% to 16.52%** and cutting RMSE **43%**.
- Built **FastAPI** service with PostgreSQL and Redis caching, containerized with **Docker** for cloud deployment.

PolicyPilot | **Medical Benefit Policy Intelligence Portal** [React, TS, Node, Gemini, MCP] May 2026

- Architected evidence-grounded **RAG pipeline** over Google Gemini 2.5-flash, forcing citation-backed answers to eliminate hallucination in high-stakes healthcare eligibility decisions.

EXTRACURRICULAR ACTIVITIES

- **Finalist, RevolutionUC 2025 Hackathon** (300 participants, 31 schools) | **SAFe Practitioner 5.1** (Mar 2023).